
IOCGuard

Mini Secure CLI Tool with Quality Checks

Module: Programming with Rust

Master in Cybersecurity

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1. Security Problem

Security analysts receive domain indicators from multiple sources (email gateways, proxy logs, manual feeds). Manually validating and classifying these is error-prone. IOCGuard automates this: it reads domains from CSV, normalizes/validates them, detects suspicious patterns using simple rules, respects an allowlist, and generates structured reports (CSV or JSON).

2. Execution Environment

All development runs inside a Docker Compose environment with the rust-labs service. The container provides Rust 1.96.0 with cargo, rustfmt, and clippy.

```
student@69811877c733:/workspace/iocguard$ rustc --version && cargo --version
rustc 1.96.0 (ac68faa20 2026-05-25)
cargo 1.96.0 (30a34c682 2026-05-25)
student@69811877c733:/workspace/iocguard$
```

Figure 1: docker compose ps showing running rust-labs service

```
cargo 1.96.0 (30a34c682 2026-05-25)
student@69811877c733:/workspace/iocguard$ cargo run -- validate --input data/domains.csv --allowlist data/allowlist.txt --out report
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.04s
Running `target/debug/iocguard validate --input data/domains.csv --allowlist data/allowlist.txt --out report`
IOCGuard report
Processed lines   : 9
Valid domains    : 6
Invalid domains  : 3
Allowlisted domains: 3
Suspicious domains: 3
Report directory : report
student@69811877c733:/workspace/iocguard$
```

Figure 2: rustc --version and cargo --version inside the container

3. Architecture & Modules

The project is organized into six modules with clear responsibilities:

- main.rs - Entry point: parses CLI args via cli.rs, calls lib.rs run().
- lib.rs - Orchestrates the pipeline: read input, process, generate report.
- cli.rs - Manual argument parsing (--input, --allowlist, --out, --json).
- domain.rs - Core logic: normalization, validation, suspicion checks.
- report.rs - CSV/JSON output and summary.txt generation.
- error.rs - Custom error types: CliError, DomainError, ReportError.

Data flow: CLI args -> CSV reader -> normalize -> validate -> suspicious check -> report (CSV/JSON) + terminal summary.

4. Data Model

```
pub enum DomainStatus { Accepted, Suspicious, Rejected }

pub struct DomainRecord {
    pub original: String,
    pub normalized: Option<String>,
    pub source: String,
    pub status: DomainStatus,
    pub reason: String,
}
```

5. Domain Validation & Normalization

Normalization: trim spaces, convert to lowercase, remove trailing dot.

A domain is rejected if:

- Empty or no dot.
- Length > 253 or a label > 63 characters.
- Contains consecutive dots (bad..domain).
- A label starts/ends with a hyphen.
- Contains invalid characters (only [a-z0-9-] allowed).
- CSV line is malformed.

```
pub fn validate(domain: &str) -> Result<(), DomainError> {
    if domain.is_empty() { return Err(DomainError::Empty); }
    if !domain.contains('.') { return Err(DomainError::NoDot); }
    if domain.len() > 253 { return Err(DomainError::TooLong(..)); }
    if domain.contains("..") { return Err(DomainError::ConsecutiveDots); }
    for label in domain.split('.') {
        if label.is_empty() { return Err(DomainError::EmptyLabel); }
        if label.len() > 63 { return Err(DomainError::LabelTooLong(..)); }
        if label.starts_with('-') || label.ends_with('-') {
            return Err(DomainError::LabelHyphenEdge(..));
        }
    }
    for c in domain.chars() {
        if !c.is_ascii_lowercase() && !c.is_ascii_digit()
            && c != '.' && c != '-' {
            return Err(DomainError::InvalidCharacter(c));
        }
    }
    Ok(())
}
```

6. Suspicion Rules

A valid domain is suspicious if NOT allowlisted and any:

- Contains xn-- prefix (internationalized domain).
- TLD is zip, mov, top, xyz, or tk (abused TLDs).
- Contains keywords: login, verify, secure, update, paypa1.
- 3+ hyphens (DGA signature).

```
pub fn check_suspicious(domain: &str, allowlist: &HashSet<String>) -> Option<String> {
    if allowlist.contains(domain) { return None; }
    let mut reasons = vec![];
    if domain.contains("xn--") { reasons.push("xn--"); }
    if let Some(tld) = domain.rsplit('.').next() {
        if ["zip", "mov", "top", "xyz", "tk"].contains(&tld) {
            reasons.push("bad TLD");
        }
    }
    for kw in &["login", "verify", "secure", "update", "paypa1"] {
        if domain.contains(kw) { reasons.push(kw); break; }
    }
    if domain.chars().filter(|&c| c == '-').count() >= 3 {
        reasons.push(">=3 hyphens");
    }
}
```

```

    if reasons.is_empty() { None } else { Some(reasons.join("; ")) }
}

```

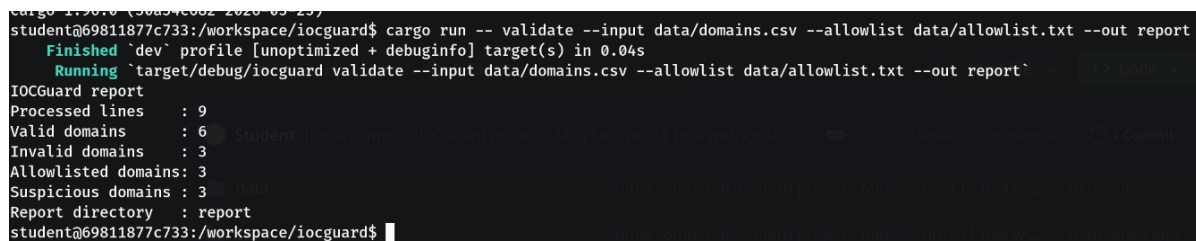
7. Command-Line Interface

```

# Standard CSV output (default)
cargo run -- validate --input data/domains.csv \
  --allowlist data/allowlist.txt --out report

# JSON output (bonus)
cargo run -- validate --input data/domains.csv \
  --allowlist data/allowlist.txt --out report --json

```



```

cargo 1.56.0 (50454e02 2020-03-23)
student@69811877c733:/workspace/iocguard$ cargo run -- validate --input data/domains.csv --allowlist data/allowlist.txt --out report
   Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.04s
   Running `target/debug/iocguard validate --input data/domains.csv --allowlist data/allowlist.txt --out report`
IOCGuard report
Processed lines      : 9
Valid domains       : 6
Invalid domains     : 3
Allowlisted domains : 3
Suspicious domains  : 3
Report directory    : report
student@69811877c733:/workspace/iocguard$

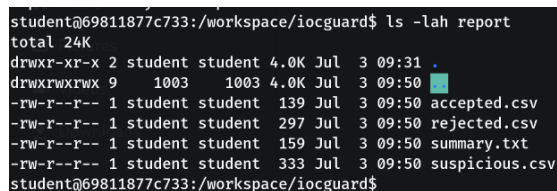
```

Figure 3: cargo run -- validate execution output

8. Generated Report Files

Output files:

- accepted.csv - Valid non-suspicious domains.
- suspicious.csv - Suspicious domains with reasons.
- rejected.csv - Invalid entries with original value and reason.
- summary.txt - Human-readable statistics.
- report.json - Structured JSON (with --json flag, bonus).



```

student@69811877c733:/workspace/iocguard$ ls -lah report
total 24K
drwxr-xr-x 2 student student 4.0K Jul  3 09:31 .
drwxrwxrwx 9 1003 1003 4.0K Jul  3 09:50 ..
-rw-r--r-- 1 student student 139 Jul  3 09:50 accepted.csv
-rw-r--r-- 1 student student 297 Jul  3 09:50 rejected.csv
-rw-r--r-- 1 student student 159 Jul  3 09:50 summary.txt
-rw-r--r-- 1 student student 333 Jul  3 09:50 suspicious.csv
student@69811877c733:/workspace/iocguard$

```

Figure 4: Generated report files (ls -lah report) and summary.txt content

9. Unit Testing

The project includes 38 tests covering validation, normalization, suspicion rules, CSV parsing, allowlist handling, and edge cases.

```

student@69811877c733:/workspace/iocguard$ cargo test
    Finished `test` profile [unoptimized + debuginfo] target(s) in 0.04s
    Running unittests src/lib.rs (target/debug/deps/iocguard-6543ba3f3d9ff0b9)

running 23 tests
test domain::tests::test_allowlisted_not_suspicious ... ok
test domain::tests::test_empty_domain ... ok
test domain::tests::test_long_domain ... ok
test domain::tests::test_load_allowlist ... ok
test domain::tests::test_long_label ... ok
test domain::tests::test_no_dot ... ok
test domain::tests::test_normalize_example_com ... ok
test domain::tests::test_normalize_trailing_dot ... ok
test domain::tests::test_normalize_trim ... ok
test domain::tests::test_parse_csv_line_missing_comma ... ok
test domain::tests::test_parse_csv_line_empty_domain ... ok
test domain::tests::test_process_line_accepted ... ok
test domain::tests::test_parse_csv_line_valid ... ok
test domain::tests::test_process_line_rejected ... ok
test domain::tests::test_process_line_suspicious ... ok
test domain::tests::test_reject_consecutive_dots ... ok
test domain::tests::test_suspicious_three_hyphens ... ok
test domain::tests::test_reject_hyphen_prefix ... ok
test domain::tests::test_suspicious_tld ... ok
test domain::tests::test_suspicious_xn ... ok
test domain::tests::test_suspicious_keywords ... ok
test domain::tests::test_suspicious_valid_but_suspicious ... ok
test tests::test_run_with_sample_data ... ok

test result: ok. 23 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

    Running unittests src/main.rs (target/debug/deps/iocguard-83d392737852f636)

```

Figure 5: cargo test successful output

10. Code Quality

All quality checks pass without errors or warnings.

```

student@69811877c733:/workspace/iocguard$ cargo fmt --check
student@69811877c733:/workspace/iocguard$ cargo fmt --check
student@69811877c733:/workspace/iocguard$ cargo clippy -- -D warnings
    Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.12s
student@69811877c733:/workspace/iocguard$

```

Figure 6: cargo fmt --check and cargo clippy -- -D warnings

10. Quality Commands

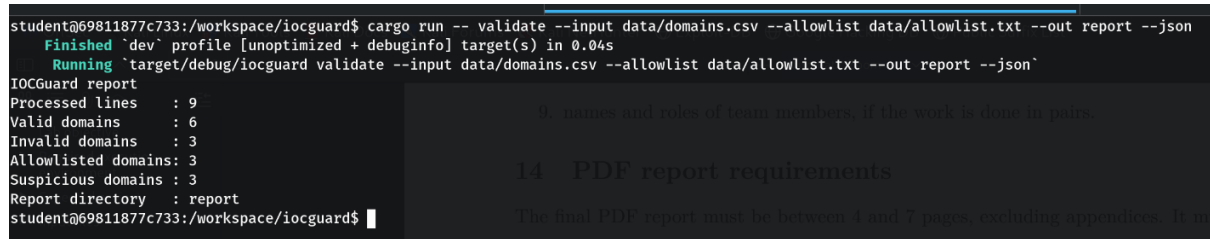
\$ cargo fmt --check	# No output = passing
\$ cargo clippy --all-targets -- -D warnings	# Zero warnings
\$ cargo test	# 38/38 tests passed

11. Bonus Features

Three bonus features were implemented for additional credit.

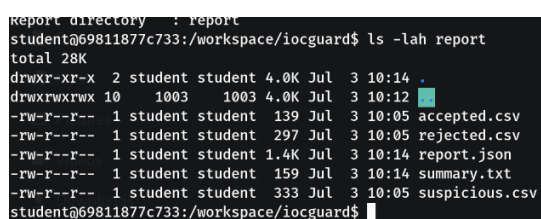
11.1 JSON Output Format (--json flag)

When --json is provided, report.json is generated instead of CSV files. It contains structured summary, accepted, suspicious, and rejected data. Allows further processing by external tools.



```
student@69811877c733:/workspace/iocguard$ cargo run -- validate --input data/domains.csv --allowlist data/allowlist.txt --out report --json
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.04s
Running `target/debug/iocguard validate --input data/domains.csv --allowlist data/allowlist.txt --out report --json`
IOCGuard report
Processed lines      : 9
Valid domains       : 6
Invalid domains     : 3
Allowlisted domains : 3
Suspicious domains  : 3
Report directory    : report
student@69811877c733:/workspace/iocguard$
```

Figure 7: cargo run with --json flag showing JSON output



```
report directory : report
student@69811877c733:/workspace/iocguard$ ls -lah report
total 28K
drwxr-xr-x  2 student student 4.0K Jul  3 10:14 .
drwxrwxrwx 10 1003      1003 4.0K Jul  3 10:12 ..
-rw-r--r--  1 student student 139 Jul  3 10:05 accepted.csv
-rw-r--r--  1 student student 297 Jul  3 10:05 rejected.csv
-rw-r--r--  1 student student 1.4K Jul  3 10:14 report.json
-rw-r--r--  1 student student 159 Jul  3 10:14 summary.txt
-rw-r--r--  1 student student 333 Jul  3 10:05 suspicious.csv
student@69811877c733:/workspace/iocguard$
```

Figure 8: JSON report file content (report.json)

11.2 Custom Error Types

Three custom error enums replace generic String errors:

- CliError: missing/unknown arguments, usage errors.
- DomainError: Empty, NoDot, TooLong, ConsecutiveDots, EmptyLabel, LabelTooLong, LabelHyphenEdge, InvalidCharacter, CsvParseError.
- ReportError: file/directory operation failures.

Each implements Display with clear messages.

11.3 Additional Edge Case Tests

12 extra tests added (total 38): underscores, spaces, uppercase, trailing dots, multiple consecutive dots, boundary lengths (253/254 chars), all suspicious TLDs and keywords tested individually.

```
test domain::tests::test_long_domain ... ok
test domain::tests::test_hyphen_suffix_label ... ok
test domain::tests::test_multiple_consecutive_dots ... ok
test domain::tests::test_long_label ... ok
test domain::tests::test_no_dot ... ok
test domain::tests::test_normalize_example_com ... ok
test domain::tests::test_normalize_trailing_dot ... ok
test domain::tests::test_normalize_trim ... ok
test domain::tests::test_parse_csv_line_missing_comma ... ok
test domain::tests::test_process_line_accepted ... ok
test domain::tests::test_process_line_rejected ... ok
test domain::tests::test_parse_csv_line_empty_domain ... ok
test domain::tests::test_parse_csv_line_valid ... ok
test domain::tests::test_process_line_suspicious ... ok
test domain::tests::test_reject_consecutive_dots ... ok
test domain::tests::test_reject_hyphen_prefix ... ok
test domain::tests::test_suspicious_keywords ... ok
test domain::tests::test_suspicious_three_hyphens ... ok
test domain::tests::test_suspicious_tld ... ok
test domain::tests::test_suspicious_valid_but_suspicious ... ok
test domain::tests::test_suspicious_xn ... ok
test domain::tests::test_trailing_dot_valid ... ok
test tests::test_run_with_sample_data ... ok
test tests::test_run_with_json_output ... ok

test result: ok. 35 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s

Running unittests src/main.rs (target/debug/deps/iocguard-83d392737852f636)

running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

Running tests/integration_tests.rs (target/debug/deps/integration_tests-d57740de9c513051)

running 3 tests
test test_without_allowlist ... ok
test test_full_pipeline ... ok
test test_json_output ... ok

test result: ok. 3 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
```

Figure 9: cargo test showing all 38 tests passing (including bonus edge cases)

12. Generated Output Examples

accepted.csv:

```
normalized_domain,source,status,reason
example.com>manual,accepted,
safe-school.edu>manual,accepted,
updates.example.org,dns_log,accepted,
```

suspicious.csv:

```
normalized_domain,source,status,reason
login-example.com,proxy_log,suspicious,keyword 'login'
xn--phishing-test.com,email_gateway,suspicious,xn--; 3 hyphens
paypa1-login.xyz,email_gateway,suspicious,.xyz TLD; 'login'
```

rejected.csv:

```
original_value,source,status,reason
bad..domain,proxy_log,rejected,consecutive dots
-malformed.com>manual,rejected,hyphen at start
verylonglabel(>63),dns_log,rejected,label > 63 chars
```

summary.txt:

```
IOCGuard Summary
Total lines processed: 9
Valid domains:        3
Invalid domains:      3
Suspicious domains:   3
Allowlisted domains:  3
```

13. Limitations & Improvements

Limitations:

- Simple suspicion rules (not production-grade threat intelligence).
- Basic CSV parsing (no quoted field support).
- Only validate subcommand implemented.
- No external crate dependencies (by design).

Future improvements:

- Add cargo audit for dependency scanning (if external crates are used).
- Benchmark with hyperfine.
- Support URL/IP indicators.
- Confidence scoring for suspicious domains.